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SWITCHING AI-BASED AUDIO PROCESSING BETWEEN TX AND RX CHANNELS

Abstract

Herein is presented a system for processing a targeted signal. The system includes a first audio signal input configured to provide a first audio signal; a second audio signal input configured to provide a second audio signal; at least one switching device coupled to the first audio signal input and the second audio signal input, and configured to provide the targeted signal to the processing circuit, the targeted signal being based on the first audio signal or the second audio signal; a processing circuit coupled to the at least one switching device and configured to process the targeted signal; and at least one controller configured to determine a state of the at least one switching device based on one or more of the first audio signal and the second audio signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/561,498, titled SWITCHING AI-BASED AUDIO PROCESSING BETWEEN TX AND RX CHANNELS, filed on Mar. 5, 2024, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

1. Field of the Disclosure

[0002] At least one example in accordance with the present disclosure relates generally to audio processing for edge device.

2. Discussion of Related Art

[0003] Electrical signals may be processed to change the signal, for example, by removing and/or attenuating unwanted frequency components, or providing gain to desirable frequency components of the signal.

SUMMARY

[0004] According to at least one aspect of the present disclosure a system for processing a targeted signal is presented, the system comprising: a first audio signal input configured to provide a first audio signal; a second audio signal input configured to provide a second audio signal; at least one switching device coupled to the first audio signal input and the second audio signal input, and configured to provide the targeted signal to the processing circuit, the targeted signal being based on the first audio signal or the second audio signal; a processing circuit coupled to the at least one switching device and configured to process the targeted signal; and at least one controller configured to determine a state of the at least one switching device based on one or more of the first audio signal and the second audio signal.

[0005] In some examples, processing the targeted signal includes using artificial intelligence-based noise suppression. In some examples, the system further comprises at least one sensor configured to provide data to the at least one controller. In some examples, the data includes one or more of voice data, energy data, background noise data, and motion data. In some examples the first audio signal input is a microphone configured to sense a sound and convert the sound into the first audio signal. In some examples, the second audio signal input is a receiver configured to receive the second audio signal. In some examples the at least one switching device includes a first switching device coupled between the processing circuit and the first audio signal input and a second switching device coupled between the processing circuit and the second audio signal input, the at least one controller being configured to control respective states of both the first switching device and the second switching device.

[0006] According to at least one aspect of the present disclosure, a system for processing audio signals is presented, comprising: a transceiver; an input-output interface; one or more switching devices coupled to the transceiver and coupled to the input-output interface; a processing circuit coupled to the one or more switching devices; and at least one controller coupled to the processing circuit and coupled to the one or more switching devices and configured to: detect an RX signal received at the transceiver, detect a TX signal received at the input-output interface, and based on the RX signal and the TX signal, control the one or more switching devices to route one of the TX signal or the RX signal to the processing circuit and route the other of the TX signal or RX signal to the transceiver or input-output interface.

[0007] In some examples, the transceiver includes a receiver configured to receive RX audio

signals, and a transmitter configured to transmit TX audio signals. In some examples, the input-output interface includes a microphone configured to convert sound into TX audio signals, and one or more speakers configured to convert RX audio signals into sound. In some examples, the transmitter is coupled to a first switching device of the one or more switching devices, and the receiver is coupled to the first switching device. In some examples, the microphone is coupled to a second switching device of the one or more switching devices, and the one or more speakers are coupled to the second switching device. In some examples, the first switching device is coupled to the second switching device. In some examples, the first switching device is configured to selectively couple the transmitter and receiver to the processing circuit and second switching device. In some examples, the second switching device is configured to selectively couple the microphone and the one or more speakers to the processing circuit and first switching device. In some examples, the at least one controller is further configured to control a state of the one or more switching devices, including the first switching device and the second switching device, based on one or more of the RX audio signals or the TX audio signals. In some examples, the system further comprises at least one sensor configured to provide data to the at least one controller.

[0008] According to at least one aspect of the present disclosure, a method of processing an audio signal is presented, comprising: receiving one or more audio signals; responsive to receiving the one or more audio signals, determining an audio signal of the one or more audio signals to be a targeted audio signal, the targeted audio signal being an audio signal of the one or more audio signals that corresponds to a most important audio signal among the one or more audio signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

[0010] FIG. 1 illustrates an audio system according to an example;

[0011] FIG. 2 illustrates an audio system according to an example;

[0012] FIG. 3 illustrates an audio system according to an example;

[0013] FIG. 4 illustrates an audio system according to an example; and

[0014] FIG. 5 illustrates a process for audio processing according to an example.

DETAILED DESCRIPTION

[0015] In many telecommunication applications, such as telephonic communications, internet communications, and so forth, audio signals transmitted from a source to a destination may include background noise. For example, in many VoIP conversations in populated areas, the sounds of other people, vehicles, machines, or other things in the area, may be transmitted along with the audio that the user intends to transmit (e.g., the user's voice).

[0016] Various audio and/or signal processing techniques may be used to adjust or eliminate background noise in signals. For example, artificial intelligence based processing techniques may be used to adjust the audio signal transmitted from the source to the destination. For example, artificial intelligence techniques may be used to provide noise suppression to the audio signal. Artificial intelligence noise suppression (AINS) techniques may therefore be used to attenuate

and/or remove the unwanted noises (such as the sounds of other people, vehicles, machines, and so forth) so that the wanted elements of the audio signal are more prominent in the signal or are the only parts of the signal that are transmitted to the destination.

[0017] Edge devices, which are typically battery-powered, wearable devices with limited power, size, and performance, may lack the capability to use AINS techniques on multiple signals simultaneously.

[0018] Aspects and elements of this disclosure relate to using AI-based techniques, including AINS, on edge devices in an efficient and effective manner. In examples herein, a half-duplex approach may be used, so that the AINS techniques (or other AI-based techniques) may be applied only to one signal at a time. In examples herein, additional techniques may be used which allow the edge device to identify an important signal, the important signal being the signal which should have an AI-based technique (such as AINS) applied to it.

[0019] For example, suppose an end-user (“user”) has a headset device with limited processing power. The headset device may have an audio output, such as a speaker, and an audio input, such as a microphone. As a result, at any given time, the user may be receiving one or more signals (a received signal may be referred to as an “RX signal” or “RX stream”) and transmitting one or more signals (a transmitted signal may be referred to as a “TX signal” or a “TX stream”). The headset may identify, from among the myriad TX and RX signals, a specific signal to be adjusted, for example, by removing from the specific signal unwanted background noise using AINS techniques.

[0020] The signal that received processing may be referred to as the “targeted signal,” as said signal is being targeted by the device for processing (e.g., via AINS or other techniques). The targeted signal may be determined and/or changed dynamically by the device. In some examples, sensors may be used to detect which signal should be the targeted signal. In some examples, computational techniques (such as statistical techniques, AI-based techniques, analysis techniques, and so forth) may be used to determine the targeted signal. In some examples, combinations of approaches (e.g., sensors and non-sensor techniques) may be used.

[0021] FIG. 1 illustrates a system **100** for applying processing to a targeted signal according to an example. The system **100** includes a headset microphone **102**, a first link **104**, a transmitter **106**, a receiver **108**, a second link **110**, and headset speakers **112**.

[0022] The system **100** is configured to receive and transmit signals. For example, the headset microphone **102** and headset speakers **112** may be parts of the same headset. When a signal is transmitted, the TX signal may originate with the headset microphone **102** and be provided via the first link **104** to the transmitter **106**. The transmitter **106** may then transmit the TX signal to a different device (e.g., a host). When the headset is receiving a signal, the RX signal may be received (e.g., from the host) at the receiver **108** and transmitted via the second link **110** to the headset speakers **112**. In some examples, the transmitter **106** and receiver **108** may be part of the same headset as the headset microphone **102** and the headset speakers **112**. In some examples, the transmitter **106** and receiver **108** may be one device (e.g., a transceiver). In some examples, there may be one or more TX signals and/or one or more RX signals.

[0023] The system **100** may send and receive signals sequentially and/or simultaneously. The system **100** may apply processing to the TX and RX signals to adjust those signals. In some examples, the processing may include AI-based techniques, such as AINS. The headset may not be able to apply AI-based techniques (or processing, more generally) to both TX and RX signals simultaneously. The system **100** may be configured to identify a targeted signal and from among the various TX and RX signals and apply processing (e.g., AINS or other AI-based techniques) to only the targeted signal.

[0024] FIG. 2 illustrates a system **200** for applying processing to a targeted signal according to an example. The system **200** includes a headset microphone **202**, a transmitter **204**, a receiver **206**, headset speakers **208**, a first switch **210**, a processing circuit **212**, a second switch **214**, and at least one controller **216** (“controller **216**”).

[0025] The system **200** is configured to apply processing to a targeted signal. The headset microphone **202** is coupled to the first switch **210**. The receiver **206** is coupled to the first switch **210**. The first switch **210** is coupled to the processing circuit **212** and the second switch **214**. The processing circuit **212** is coupled to the first switch **210** and the second switch **214**. The second switch **214** is coupled to the transmitter **204** and to the headset speakers **208**. The controller **216** may be communicatively coupled the first switch **210** and to the second switch **214**, and may additionally be communicatively coupled to the processing circuit **212**, the transmitter **204**, the receiver **206**, the headset microphone **202**, and/or the headset speakers **208**.

[0026] There are at least two potential processing paths in the system **200**. A first path corresponds to processing a TX signal originating from the headset microphone **202** and being transmitted to another device via the transmitter **204**. A second path corresponds to processing an RX signal received at the receiver **206**, provided to and output by the headset speakers **208**. TX and RX signals may be present in the system **200** simultaneously or sequentially. In some examples, sequences of RX and TX signals may rapidly alternate between TX and RX signals (e.g., an RX signal may be received at a first time, a TX signal at a second time shortly later, and another RX signal at a third time shortly after the first time or the second time). The switches **210**, **214** route a targeted signal to the processing circuit **212**, and may allow other signals to bypass the processing circuit **212**, thereby reserving the resources of the processing circuit **212** for the targeted signal while still allowing receipt or transmission of other signals.

[0027] The headset microphone **202** is configured to receive sounds and convert those sounds into electrical signals (for example, the TX signal), and then to provide the TX signal to the first switch **210**. The headset microphone **202** may contain, for example, a diaphragm, magnet, coil, and so forth, that are configured to convert sounds into electrical signals.

[0028] The receiver **206** is configured to receive a signal (for example, the RX signal) from another device and to provide the RX signal to the first switch **210**. The RX signal received by the receiver **206** may be, for example, a signal corresponding to a desired sound to be output by the headset speakers **208**.

[0029] The transmitter **204** is configured to receive a signal (e.g., the TX signal) and transmit the TX signal to another device, such as a computer, phone, radio, and so forth.

[0030] The headset speakers **208** are configured to receive an electrical signal (e.g., the RX signal) and convert the RX signal into sound. The headset speakers **208** may contain, for example, a diaphragm, magnet, coil, and so forth, that are configured to convert electrical signals into sound. The headset speakers **208** may be configured to sit in or near the ear of a user so that the user may hear the sound generated based on the signal (e.g., based on the RX signal). The headset speakers **208** may contain amplifiers or other circuitry configured to adjust the gain and/or attenuation of the RX signal (e.g., to make the RX signal louder or quieter).

[0031] The first switch **210** is configured to select between the headset microphone **202** and the receiver **206** and route the signals received from one of those two components to the processing circuit **212** at a time. For example, the headset microphone **202** provides TX signals and the receiver **206** provides RX signals to the first switch **210**. Thus, the first switch **210** may route one of the TX signals or the RX signals to the processing circuit **212** (the signal provided to the processing circuit **212** therefore being the targeted signal). The first switch **210** may be further configured to directly route whichever of the TX or RX signals is not the targeted signal to the second switch **214**, thereby bypassing the processing circuit **212**. The signal that bypasses the processing circuit **212** may be referred to hereafter as the bypassed signal.

[0032] The first switch **210** may have a first state corresponding to providing the TX signal to the processing circuit **212** (e.g., as the targeted signal) and/or providing the RX signal to the second switch **214** (e.g., as the bypassed signal). The first switch **210** may have a second state corresponding to providing the RX signal to the processing circuit **212** (e.g., as the targeted signal) and/or providing the TX signal to the second switch **214** (e.g., as the bypassed signal).

[0033] The second switch **214** is configured to receive the targeted signal and/or the bypassed signal and route those signals to the transmitter **204** and the headset speakers **208**. In particular, the second switch **214** may be configured to route whichever of the bypassed signal and the targeted signal corresponds to the TX signal to the transmitter **204**, and whichever of the bypassed signal and the targeted signal corresponds to the RX signal to the headset speakers **208**.

[0034] The second switch **214** may have a first state corresponding to providing the targeted signal to the transmitter **204** and/or the bypassed signal to the headset speakers **208**. The second switch may have a second state corresponding to providing the bypassed signal to the transmitter **204** and/or the targeted signal to the headset speakers **208**.

[0035] The processing circuit **212** applies processing to the targeted signal. The processing applied may include AI-based techniques, such as AINS. The processing may remove or attenuate unwanted background noise from the targeted signal whether the targeted signal is derived from a TX signal or an RX signal. The processing may adjust the targeted signal in various other ways as well or in addition to those described above, for example by adjusting the harmonic components of the targeted signal, adjusting attenuation and/or gain for specific frequencies making up the targeted signal, adjust the speed or tone of the signal, and so forth.

[0036] The controller **216** may control the states of the switches **210**, **214**. In some examples, the controller **216** may control the processing circuit **212**, for example, by determining what sort of processing the processing circuit **212** should provide to the targeted signal. The controller **216** may determine the state of the switches **210**, **214** based on a determination of which signal should be the targeted signal. For example, if the TX signal is the targeted signal, the controller **216** may control the first switch **210** to be in the first state and the second switch **214** to be in the first state. For example, if the RX signal is the targeted signal, the controller **216** may control the first switch **210** to be in the second state and the second switch **214** to be in the second state.

[0037] The controller **216** may determine which signal should be the targeted signal in various ways. In some examples, the controller **216** may use a sensor input to determine the targeted signal. For example, if the headset microphone **202** includes a sensor configured to detect motion of the user's jaw (e.g., via conduction, movement detection, and so forth), the controller **216** may determine that the user is talking and therefore that the TX signal should be the targeted signal. In some examples, if the user is not talking (e.g., motion of the jaw is not detected), the controller **216** may determine that the TX signal should not be the targeted signal (and may default to considering the RX signal to be the targeted signal). In some examples, the controller **216** may determine which signal should be the targeted signal based on the duration of the signal (e.g., the controller **216** may favor signals with longer or shorter durations, or may favor continuous signals over discontinuous signals). In some examples, the controller **216** may determine which signal should be the targeted signal based on another metric, such as the relative power of the signals (e.g., power spectral density), whether a signal contains frequency components corresponding to the human voice, whether the user has designated a particular signal for processing, and so forth. In some examples, the controller **216** may use an AI-algorithm to identify which signal should be the targeted signal.

[0038] FIG. 3 illustrates a system **300** for providing processing to a targeted signal according to an example. The system **300** is similar to the system **200** and shows the processing paths in an explicitly separated configuration.

[0039] The system **300** includes a first processing path **302** (“first path **302**”), a headset microphone **304**, a first processing circuit **306**, a transmitter **308**, a second processing path **310** (“second path **310**”), a receiver **312**, a second processing circuit **314**, and headset speakers **316**.

[0040] The headset microphone **304** and headset speakers **316** may be part of the same headset. The transmitter **308** and receiver **312** may be part of the same headset and/or may be part of the same circuit (e.g., the same transceiver). The first processing circuit **306** and second processing circuit **314** may be part of the same headset and/or may be part of the same circuit (e.g., the same processing circuit).

[0041] The first path **302** corresponds to processing of a TX signal. The second path **310** corresponds to processing of an RX signal.

[0042] The headset microphone **304** is configured to convert sound into a TX signal and to provide the TX signal to the first processing circuit **306**. The first processing circuit **306** is configured to process the TX signal (e.g., by applying AI-based techniques, such as AINS), or not (e.g., to simply bypass the TX signal). The first processing circuit **306** is further configured to provide either the processed TX signal (e.g., the targeted signal) or the bypassed TX signal (e.g., the bypassed signal) to the transmitter **308**. The transmitter **308** is configured to transmit the TX signal (either as a bypassed signal if the TX signal was not processed or as the targeted signal if the TX signal was processed) and provide the TX signal to another device (e.g., via wireless or wired transmission).

[0043] The receiver **312** is configured to receive an RX signal from another device (e.g., via wireless or wired transmission), and to provide the RX signal to the second processing circuit **314**. The second processing circuit **314** is configured to either process the RX signal (e.g., by applying AI-based techniques, such as AINS) or not (e.g., to simply bypass the RX signal). The second processing circuit **314** is further configured to provide either the processed RX signal (e.g., the targeted signal) or the bypassed RX signal (e.g., the bypassed signal) to the headset speakers **316**. The headset speakers **316** are configured to receive the RX signal (either as the bypassed signal if the RX signal was not processed or as the targeted signal if the RX signal was processed), and convert the RX signal from electrical form into sound (e.g., sound audible to the user of the headset).

[0044] In some examples, the system **200** may require less computation time compared to the system **300**, and the battery life of the system **200** may be greater than that of system **300** (e.g., the system **200** may use less power than that of system **300**).

[0045] FIG. **4** illustrates an audio system **400** according to an example. The audio system **400** is, in some example, an edge device, such as a headset. The audio system **400** includes a transceiver **402**, a processing circuit **410**, a first routing device **408**, a second routing device **412**, an input and output interface **414** (“IO interface **414**”), and at least one controller **420** (“controller **420**”). The transceiver **402** includes a transmitter **404** and a receiver **406**. The IO interface **414** includes a microphone **416** and one or more speakers **418** (“speakers **418**”). The audio system **400** may also include one or more optional sensors **422** (“sensor **422**”).

[0046] The transceiver **402** is coupled to the first routing device **408**. In some examples, the transmitter **404** is coupled to a first connection of the first routing device **408** and the receiver **406** is coupled to a second connection of the first routing device **408**. The first routing device **408** is further coupled (via a third connection) to the processing circuit **410** and via a fourth connection to the second routing device **412**. The processing circuit **410** is coupled to the first routing device **408** and to the second routing device **412**. The second routing device **412** is further coupled to the IO interface **414**. In some examples, the second routing device **412** is coupled to the microphone **416** via a first connection and to the speakers **418** via a second connection. The controller **420** is communicatively coupled to the first routing device **408**, the second routing device **412**, and to the processing circuit **410**. The controller **420** may also be coupled to the transceiver **402** and/or IO interface **414**, and may be individually coupled to any constituent elements of the transceiver **402** and/or IO interface **414**. The sensor **422** may be coupled to the controller **420**.

[0047] The transceiver **402** is configured to send and receive signals (e.g., wirelessly or via a wired connection) to and/or from the audio system **400** and at least one other device (such as a computer). The transmitter **404** may, for example, transmit a TX signal originating with the audio system **400** to one or more other devices. The receiver **406** may, for example, receive an RX signal originating with one or more other devices, and provide the RX signal to the audio system **400**.

[0048] The IO interface **414** is configured to provide RX signals as sound to the user and to convert sound into TX signals from the user. The microphone **416** may convert sound (e.g., from the user or the space around the user) into electrical signals for use by the audio system **400**. The speakers **418**

may convert electrical signals within the audio system **400** into audible sound.

[0049] The first routing device **408** is configured to route RX signals from the receiver **406** to the processing circuit **410** or to the second routing device **412**, depending on whether the RX signal is the targeted signal. The first routing device **408** is further configured to route TX signals from the processing circuit **410** or from the second routing device **412** to the transmitter **404** depending on whether the TX signal is the targeted signal. The first routing device **408** may include one or more switches to control the routing.

[0050] The second routing device **412** is configured to route RX signals from the processing circuit **410** or the first routing device **408** to the speakers **418**, depending on whether the RX signal is the targeted signal. The second routing device **412** is further configured to route TX signals from the microphone **416** to the processing circuit **410** or the first routing device **408**, depending on whether the TX signal is the targeted signal. The second routing device **412** may include one or more switches to control the routing.

[0051] When a signal is not a targeted signal (e.g., the signal is a bypassed signal), the routing devices **408**, **412** may provide the signal directly to one another. For example, when a TX signal is the bypassed signal, the second routing device **412** may provide the TX signal directly to the first routing device **408**. When an RX signal is the bypassed signal, the first routing device **408** may provide the RX signal directly to the second routing device **412**.

[0052] The processing circuit **410** is configured to process the targeted signal (e.g., a TX or RX signal) and provide the targeted signal to one of the routing devices **408**, **412**. When a TX signal is the targeted signal, the processing circuit **410** may provide the targeted signal to the first routing device **408**. When an RX signal is the targeted signal, the processing circuit **410** may provide the targeted signal to the second routing device **412**.

[0053] The sensor **422** is configured to provide data to the controller **420**. For example, the sensor may be a motion sensor, a bone conduction sensor, a sound sensor, an energy sensor, and so forth. The sensor **422** may therefore provide information indicative of the motion of the user, the speaking of the user, the sounds made by the user and/or around the user, and/or the energy of sounds made by the user and/or around the user. In some examples, the sensor **422** and/or audio system **400** may include voice activity detections systems, such as those disclosed in U.S. Pat. No. 11,614,916, titled “User Voice Activity Detection,” which is hereby incorporated by reference in its entirety.

[0054] The controller **420** is configured to control the system **400** and to determine which signal (among all available signals, e.g., all TX and RX signals) should be the targeted signal. The controller **420** may control the state of the first routing device **408** and/or second routing device **412**. For example, the controller **420** may determine whether the first routing device **408** has connected the transmitter **404** or the receiver **406** to the processing circuit **410** and/or whether the first routing device **408** has connected the receiver **406** and/or transmitter **404** to the second routing device **412** (e.g., bypassing the processing circuit **410**). Similarly, the controller **420** may determine whether the second routing device **412** has connected the microphone **416** and/or the speakers **418** to the processing circuit **410** and/or whether the second routing device **412** has connected the speakers **418** and/or the microphone **416** to the first routing device **408** (e.g., bypassing the processing circuit **410**).

[0055] FIG. 5 illustrates a process **500** for processing a targeted signal according to an example.

[0056] At act **502**, a controller (such as the controller **420**) determines how many signals are being received. If the controller determines that only one signal is being received (**502** NO), whether the signal is a TX signal or an RX signal, the process **500** may continue to act **506**. If the controller determines that more than one signal is being received (e.g., two or more signals, whether those signals are RX signals, TX signals, or a mix of RX and TX signals) (**502** YES), the process **500** may then continue to act **504**.

[0057] At act **504**, the controller determines which signal among the multiple signals should be the

targeted signal, and thus should be processed using the limited resources of the edge device (e.g., the headset). The controller may determine which signal should be the targeted signal using an AI-based technique, such as an AI-algorithm, or another technique. For example, the controller may use the energy of one or more signals, the length or duration of one or more signals, an input from one or more sensors (such as a sensor **422**), and so forth. The process **500** may then proceed to act **506**.

[0058] At act **506**, the targeted signal is provided to the processing circuit (e.g., the processing circuit **410**). The processing circuit then processes the targeted signal (e.g., using AINS or another technique).

[0059] In some examples above, the system (e.g., system **100, 200, 300, 400**) may feather the processing (e.g., the AI-based processing) when applying said processing to a given signal. For example, rather than fully process an audio signal in a manner that might be abrupt and potentially noticeable to a user, the system can instead begin with relatively less processing of the signal and increase the processing over time (e.g., begin with relatively little attenuation of the background noise and increase the attenuation over time). If the system is able to predict or know when the audio signal will terminate, similar feathering may be applied during termination of the signal, such that the amount of processing is reduced as the signal approaches termination. For example, during the middle of a transmission the attenuation of background noise may be high, but as the signal reaches an end the attenuation may be reduced over time or completely eliminated.

[0060] The foregoing examples apply to any type of electrical signal, and need not be limited to purely audio signals.

[0061] Various controllers, such as the controller **420**, may execute various operations discussed above. Using data stored in associated memory and/or storage, the controller **420** also executes one or more instructions stored on one or more non-transitory computer-readable media, which the controller **420** may include and/or be coupled to, that may result in manipulated data. In some examples, the controller **420** may include one or more processors or other types of controllers. In one example, the controller **420** is or includes at least one processor. In another example, the controller **420** performs at least a portion of the operations discussed above using an application-specific integrated circuit tailored to perform particular operations in addition to, or in lieu of, a general-purpose processor. As illustrated by these examples, examples in accordance with the present disclosure may perform the operations described herein using many specific combinations of hardware and software and the disclosure is not limited to any particular combination of hardware and software components. Examples of the disclosure may include a computer-program product configured to execute methods, processes, and/or operations discussed above. The computer-program product may be, or include, one or more controllers and/or processors configured to execute instructions to perform methods, processes, and/or operations discussed above.

[0062] Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

[0063] Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are not intended to limit the presently disclosed systems or methods,

their components, acts, or elements. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

[0064] References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for irreconcilable differences, the term usage in this document controls.

[0065] Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of, and within the spirit and scope of, this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A system for processing a targeted signal, comprising: a first audio signal input configured to provide a first audio signal; a second audio signal input configured to provide a second audio signal; at least one switching device coupled to the first audio signal input and the second audio signal input, and configured to provide the targeted signal to the processing circuit, the targeted signal being based on the first audio signal or the second audio signal; a processing circuit coupled to the at least one switching device and configured to process the targeted signal; and at least one controller configured to determine a state of the at least one switching device based on one or more of the first audio signal and the second audio signal.
2. The system of claim 1 wherein processing the targeted signal includes using artificial intelligence-based noise suppression.
3. The system of claim 1 further comprising at least one sensor configured to provide data to the at least one controller.
4. The system of claim 3 wherein the data includes one or more of voice data, energy data, background noise data, and motion data.
5. The system of claim 1 wherein the first audio signal input is a microphone configured to sense a sound and convert the sound into the first audio signal.
6. The system of claim 1 wherein the second audio signal input is a receiver configured to receive the second audio signal.
7. The system of claim 1 wherein the at least one switching device includes a first switching device coupled between the processing circuit and the first audio signal input and a second switching device coupled between the processing circuit and the second audio signal input, the at least one controller being configured to control respective states of both the first switching device and the second switching device.
8. A system for processing audio signals, comprising: a transceiver; an input-output interface; one or more switching devices coupled to the transceiver and coupled to the input-output interface; a processing circuit coupled to the one or more switching devices; and at least one controller coupled to the processing circuit and coupled to the one or more switching devices and configured to: detect an RX signal received at the transceiver, detect a TX signal received at the input-output interface, and based on the RX signal and the TX signal, control the one or more switching devices to route one of the TX signal or the RX signal to the processing circuit and route the other of the TX signal or RX signal to the transceiver or input-output interface.
9. The system of claim 8 wherein the transceiver includes a receiver configured to receive RX audio signals, and a transmitter configured to transmit TX audio signals.
10. The system of claim 9 wherein the input-output interface includes a microphone configured to

convert sound into TX audio signals, and one or more speakers configured to convert RX audio signals into sound.

- 11.** The system of claim 10 wherein the transmitter is coupled to a first switching device of the one or more switching devices, and the receiver is coupled to the first switching device.
 - 12.** The system of claim 11 wherein the microphone is coupled to a second switching device of the one or more switching devices, and the one or more speakers are coupled to the second switching device.
 - 13.** The system of claim 12 wherein the first switching device is coupled to the second switching device.
 - 14.** The system of claim 13 wherein the first switching device is configured to selectively couple the transmitter and receiver to the processing circuit and second switching device.
 - 15.** The system of claim 14 wherein the second switching device is configured to selectively couple the microphone and the one or more speakers to the processing circuit and first switching device.
 - 16.** The system of claim 15 wherein the at least one controller is further configured to control a state of the one or more switching devices, including the first switching device and the second switching device, based on one or more of the RX audio signals or the TX audio signals.
 - 17.** The system of claim 8 further comprising at least one sensor configured to provide data to the at least one controller.
 - 18.** A method of processing an audio signal, comprising: receiving two or more audio signals; determining a first audio signal of the two or more audio signals to be a targeted audio signal, the targeted audio signal being an audio signal of the one or more audio signals that corresponds to a most important audio signal among the one or more audio signals; determining at least one second audio signal of the two or more audio signals to be a bypassed audio signal, the bypassed audio signal being an audio signal that is less important than the targeted audio signal; forwarding the bypassed audio signal to a transmitter or receiver; and processing the targeted audio signal to change at least one characteristic of the targeted audio signal.
 - 19.** The method of claim 18 further comprising determining which audio signal of the two or more audio signals is the most important audio signal based on at least one element of a set, the set including: motion of a user's jaw, duration of the audio signal, relative power of the audio signal with respect to power of other audio signals of the two or more audio signals, or frequency components of the audio signal.
 - 20.** The method of claim 18 further comprising forwarding the targeted audio signal to a transmitter or receiver responsive to processing the targeted audio signal.
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